Problem Statement — Bring the Physical World to AI

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Granica × IIT Guwahati Hackathon · 48 hours · teams of 2–3

Physical AI wants its data

1. Captured data made modern AI possible

The remarkable progress in AI began with a gift: much of the raw material was already there. Books, web pages, code, images, and conversations had been captured and stored. Models learned from them. The leap from the first GPT systems to the models we use now was driven not only by new architectures, but by more data, richer data, and better ways of using it.

2. Self-driving shows what happens when the physical world is captured

The goal of self-driving was not to build a large road dataset. It was to move people safely without a human at the wheel. Teams worked backward from that problem: what must a car notice, how should roads and edge cases be recorded, and how can those observations support a decision in the next second? The data, labels, sensors, and models all existed to make the car more capable in the world. After years of work, self-driving cars are doing remarkable things. Self-driving remains one of the first physical domains to be made learnable.

At the beginning, this work would have looked much narrower than “build a self-driving car”: keep the car in its lane, detect a pedestrian, anticipate another road user’s move, or handle one route in one kind of weather. Each question was tied to a failure the system needed to avoid. That is how difficult physical problems become buildable.

3. Most physical systems are still waiting

We do not have good records of how a campus moves, how a hospital operates, where a city wastes energy, or what a factory sounds like before a failure. Models cannot help with a problem if nobody has

captured the right evidence from the workflow around it. The next generation of models that can understand and help in the physical world will need the same foundation today's models had: data gathered for a purpose.

4. Your opportunity at IIT Guwahati

In 48 hours, you will choose a real problem, name the person or team who has it, and build the smallest useful intervention you can. You might collect physical data, turn it into a structured record, and use AI to make a decision, recommendation, alert, or workflow better. You will not be asked to solve the whole world. You will learn how to build one small, credible piece of the future—and how to make a project that someone else can inspect, trust, and use.

IIT Guwahati is full of systems to work with: hostels, messes, classrooms, roads, lakes, workshops, the hospital, and people moving through all of them. You do not need a large budget or elaborate hardware. You need a good question and the patience to collect evidence.

Start with the problem, then work backward:

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real problem → physical workflow → useful observations → AI-assisted decision → action
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The point is not to collect data for its own sake or wrap a chatbot around a made-up example. It is to understand a real workflow, produce the evidence it is missing, and show what someone could do differently because you built the right support around it.

Do not start with a sensor. Start with someone who needs something to work better.

Your challenge

1. **Choose a real problem and a user.** What decision is difficult today? What action would improve if the right evidence were available?
2. **Understand the physical workflow.** What actually happens, where does it happen, and what can you observe without disrupting it?
3. **Collect the evidence.** Manual counts, photographs, phone sensors, a webcam, public APIs, or an approved campus system are all valid. Cheap is fine. Real is required.
4. **Make it usable.** Collect more than once, record how and when you collected it, and put the results in an open format. Parquet is preferred.

5. **Use AI where it earns its place.** It might filter images, classify sounds, extract information, find patterns, predict what happens next, or help a person decide what to do.

You decide how far to take the solution:

- **Understand —** fully sufficient to win. Give the user a clearer picture of what is happening and why. A new understanding that changes how someone works is a complete project.
- **Predict —** a bonus. What is likely to happen next?
- **Recommend —** a bonus. What should someone do differently?
- **Actuate —** optional, never expected. Make something respond to the result if it is safe and sensible.

Examples of problem-led projects

Each example starts with a problem, a user, and a possible action. The physical data and AI are means to that end. Do not copy them. Let them show you the shape of a good project: small enough to build in 48 hours, useful enough that someone would want to use it afterward.

- **Motion — help estates decide what to fix first.** People using wheelchairs or walking aids may know which routes are difficult, but the estates office may not have a current, usable record. Walk campus routes with a phone, mark kerbs without ramps, broken surfaces, missing handrails, and blocked paths, then turn those observations into a route map and a ranked repair list. **Who could use it:** the estates office. **What changes:** it can decide where an access improvement would help most.
- **Sight — help facilities find avoidable energy waste.** The facilities team cannot check every room by hand after the academic complex empties. Photograph the buildings late on both nights, count rooms that are still lit or cooled, and compare that with the timetable. Turn the result into a nightly list of rooms to check, with an estimate in kWh and rupees. **Who could use it:** the School of Energy Science and Engineering and the person who pays the bill. **What changes:** staff get a targeted action instead of a general warning.
- **Sound — help preserve languages that are rarely recorded.** The Centre for Linguistic Science and Technology needs labelled speech, but collecting and sorting it is slow. With clear consent, record short passages from languages represented on campus—perhaps Assamese, Bodo, Maithili, Nepali, and others you meet. Let a language-identification model make a first pass, then check the labels with speakers. **Who could use it:** the language centre. **What changes:** they receive a checked research collection they can build on.

What these projects have in common is not a fancy model. They begin with somebody who needs to do something, then use a photograph or a recording to make that action more informed. The result is a decision, a maintenance list, an energy bill, or a research collection—not just a dashboard.

Other problems worth tackling:

- helping campus security prioritise dark stretches on night routes, using where people actually walk;
- giving maintenance teams a severity-ranked list from systematic facade and road photographs;
- helping mess managers reduce contamination in waste bins or plate waste across hostels;
- making the institute's tree-canopy and carbon reporting easier to update;
- giving people a rough, local haze record when the nearest air-quality readings are far away;
- helping a workshop spot a change in machine sound before a failure.

What counts, and what does not

Allowed: manual counts and tally sheets · phone sensors (camera, microphone, GPS, accelerometer) · photographs processed by a vision model · public APIs · your own AI accounts and API keys, from any provider · local models · AI coding assistants · planning and environment setup before the event · clearly labelled synthetic data used only to extend real observations.

Not allowed: synthetic-only projects · covert recording of identifiable people · entering restricted areas or touching campus infrastructure · unauthorised access to campus systems · bringing a mostly finished project from before the event.

Five rules:

1. **Teams of 2–3.** Planning before kickoff is allowed. Substantial implementation and qualifying data collection happen during the 48 hours.
2. **Real data is required.** Synthetic data may extend your observations; it cannot replace them. Show the split.
3. **Respect privacy and safety.** Do not covertly record identifiable people. Aggregate, anonymise, or get consent. Observe only where organisers approve.
4. **Bring your own AI.** Use your own models, accounts, and keys. No compute or credits are provided. Assistants are welcome, but you must be able to explain what you submit.
5. **Make it inspectable.** Provide a data sample, a schema, and a secure way to access the code. A public repository is not required.

What you submit

#	Artefact	Format
1	**3-slide deck**	PDF. Slide 1: the problem, the user, and your question. Slide 2: the workflow, your collection method, and the solution you built. Slide 3: the result, what changes for the user, and the demo link. Hard limit of 3 slides.
2	**3-minute demo**	Live URL or recorded video. Recorded is safer.
3	**Code + data access path**	Private repository invite, archive, or a scheduled walkthrough. Include a data sample and a schema description.

Also document the row count, collection window and cadence, where observations came from, what is observed versus inferred versus synthetic, and how AI was used.

Judges review every submission. The **top 10 teams present live** to a panel that includes Granica engineers and external guest judges. Make your slides something you can stand up and explain in three minutes.

How you are judged

The judges will follow the same path as your project: the problem, the workflow, the evidence, and what changes.

Criterion	Weight	What earns the points
1. Idea	25%	The problem you picked, the person who has it, and the question you asked. Is it genuinely rooted in the world, not a screen? Is the question sharp and answerable in 48 hours? Does the proposed solution matter to someone?
2. Data collection process	25%	How you turned a real workflow into useful observations. A clever proxy can beat expensive equipment. Show your method, consistency, and what you did when collection went wrong.
3. The data collected	25%	What you actually ended up with: structure, schema, open format, traceability, and honesty about gaps and bias. Is it the right evidence for the problem? Could a stranger load the dataset and understand it without you in the room?
4. Impact	25%	What the project makes possible. Does it help the named user make a better decision or take a better action? Does the demo make that clear in three minutes?

Your demo and your code are judged inside these four criteria, not as separate line items.

You are not judged on model sophistication. A team that counts a queue by hand every 15 minutes, stores it cleanly, and draws an honest conclusion will beat a team with a sophisticated model and no real data.

Two things lose points: claiming more than your observations support, and handing over a dataset nobody else can interpret. Two things win them: a collection process still running while you present, and an honest account of what your data cannot tell you.

A full participant handbook—with worked examples, sample schemas and collection plans, a suggested free tool stack, a realistic 48-hour plan, and answers to common questions—is provided with the listing and at kickoff.